

Interview exercise: build an agent for the Tournament of Towns archive

Candidate brief and interviewer scorecard | Draft for colleague review | 26 August 2026

Format: 1 day take-home, followed by a 30-minute walkthrough and live change.

What this tests

Can the candidate build an agent that turns an inconsistent archive into reliable English PDFs and detects when it has failed?

We assess the agent, harness and translations. The workflow must be able to process the full archive and surface failures.

Candidate brief

Task

Using the [Tournament of Towns problem index](#), build an agent that can create English PDFs for every written problem set in the archive.

Organise the outputs as:

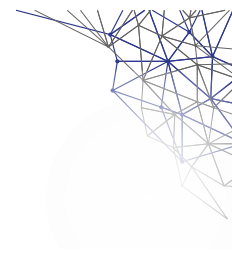
```
output/
  junior/
  basic/
  advanced/
  senior/
  basic/
  advanced/
```

Use this working classification, checking it against the source:

Source group	Output folder
8-9 classes or equivalent O-level	junior
10-11 classes or equivalent A-level	senior
Basic or базовый вариант	basic
Advanced or сложный вариант	advanced

Within the timebox, demonstrate the workflow on:

- 18 October 2009, basic, 8-9 classes;
- 18 October 2009, basic, 10-11 classes; • 25 October 2009, advanced, 8-9 classes;
- 25 October 2009, advanced, 10-11 classes.



Required approach

You may use an existing coding agent. Run it through a local harness that provides filesystem access, code execution, package installation, model access, caching, checkpoints, validation and cost tracking.

Keep the prompts or instruction files reusable and versioned. A chat transcript plus manually assembled PDFs does not count.

Constraints

- You receive an OpenRouter key with a hard \$5 spending cap. Record the model, call count and spend, and do not bypass the cap.
- Keep the key out of code, logs and PDFs.
- The four demonstration PDFs must be generated by the same workflow intended for the full corpus. Do not hand-edit the samples.
- Preserve mathematical meaning, notation, numbers, conditions, problem numbers, point values, authors and relevant diagrams.
- Exclude solutions. Record any source item that is ambiguous, duplicated or outside your chosen written-problem scope.
- You may use existing English editions as disclosed control sources. Do not present them as your translation or design. Show the model-based translation path on at least one non-English source or controlled excerpt.

Deliverables

- 1 A runnable agent and harness, including the prompts or instruction files.
 - 2 The four demonstration PDFs in the required folders.
 - 3 A manifest of the source inventory and outputs, with source URLs, classifications, problem counts, hashes and status.
 - 4 A short README explaining how to run, resume and inspect the work, including model and cost choices.
 - 5 A run trace or QA note showing what was checked, what was sampled and what remains unresolved.
- Reconcile the inventory automatically. Render every generated PDF. Inspect all four examples page by page. For each source format, compare at least one problem with the original.
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